

FORD VEV

Proving Ground Operational System

Painel de Analises - Streamlit

Documentacao Tecnica - ABNT NBR 14724

Gerado em 08/06/2026

Dashboard gerencial para coordenação e engenharia, construído em Streamlit 1.31 com Plotly para visualizações, consumindo dados do Firebase Realtime Database.

1 Access

```
http://localhost:8501
```

Start command:

```
python -m streamlit run dashboard/app.py --server.headless true
```

2 Theme

Ford Motor Co. dark theme, defined in .streamlit/config.toml:

Token	Value	Usage
`primaryColor`	`#38bdf8`	Cyan accent
`backgroundColor`	`#0f172a`	Dark navy
`secondaryBackgroundColor`	`#1e293b`	Card surfaces
`textColor`	`#f8fafc`	Light text

Two theme variants available in sidebar: Premium Dark (default) and Industrial Light (light mode).

3 10 Dashboard Tabs

3.1 Tab 0 — Resumo Gerencial

Executive overview for management. Shows:

- Operational Score (composite 0-100 with color coding)
- Traffic Light KPIs: Fuel alerts, active laudos, overdue maintenance
- Top 5 Vehicles by KM (bar chart)
- Fleet Summary: Total vehicles, active operators, fuel consumption, incident rate

3.2 Tab 1 — Visão Geral

Comprehensive fleet overview with filters (project, operator, vehicle, date range):

- Distance timeline: KM accumulated over time (line chart)
- Operator ranking: Bars by total distance
- Vehicle usage heatmap: Hours × days matrix
- KPI Cards: Total KM, total liters, avg consumption, sessions count

3.3 Tab 2 — Monitoramento Ao Vivo

Real-time active shifts:

- Active operators cards with current vehicle, timer, project
- Live fuel consumption gauge
- Active sessions timeline
- GPS breadcrumbs map (Folium) - last known positions

3.4 Tab 3 — Combustível

Fuel analytics:

- FEG trends: Km/L per vehicle over time
- Station comparison: Fuel cost/efficiency by posto
- Refueling frequency: Histogram of intervals
- Project fuel distribution: Pie chart of liters by project
- Anomaly detection: Flag outliers in consumption ($>2\sigma$ from mean)

3.5 Tab 4 — Laudos e Saúde

Technical issue (laudo) tracking:

- Issue severity breakdown: Critical/High/Medium/Low
- Affected systems pie chart: Motor, freios, suspensão, etc.
- Vehicle health score: 0-100 composite score per vehicle
- Open vs resolved trend over time
- MTBF (Mean Time Between Failures) per system

3.6 Tab 5 — Protocolos de Teste

Active test protocols and compliance:

- Protocol completion rate: % of laps/checkpoints completed
- Track time distribution: Interno vs Externo
- R389 compliance: Per-vehicle protocol adherence
- Braking/deceleration test results

3.7 Tab 6 — Ranking de Pilotos

Driver performance competitive ranking:

- Composite pilot score: Efficiency + safety + consistency
- FEG ranking: Best fuel efficiency
- KM/Lap profile: Per-pilot distance and productivity
- Multi-metric leaderboard: Sortable by distance, efficiency, sessions

3.8 Tab 7 — Analytics Avançado

Advanced statistical analysis:

- Fleet benchmarking: Vehicle vs fleet averages
- Anomaly detection: IQR-based outliers on consumption, distance, speed
- Trend forecasting: Linear regression on KM accumulation
- Correlation matrix: KM vs consumption, speed vs efficiency
- Whisker plots: Distribution of key metrics per project

3.9 Tab 8 — Explorador de Dados

Raw data explorer:

- Data source selector: Turnos, Abastecimentos, Laudos, Sessões
- Filterable table: Multi-column sort and filter
- CSV/Excel export: Download filtered data
- SQL-like query: Date range + column selection

3.10 Tab 9 — Cards de Carros

Visual vehicle cards grid (Ford dark theme):

- Vehicle card per vehicle with:
- Name, VIN, photo placeholder
- Stat circles: KM, consumption, sessions (color-coded by threshold)
- Progress bars: KM vs fleet benchmark, consumption vs target
- Health badge: 0-100 score with gradient colors
- Pilot avatar: Initials in colored circle
- Project tags: Colored project badges
- Search: By name or VIN
- Sort: By name, KM, consumption, health score
- Expand: Full details with laudos table

4 Data Loading

4.1 Firebase REST Endpoints

Endpoint	Description	Cache Strategy
`vev_turnos_encerrados.json`	Completed shifts	Session state fallback
`vev_turnos_ativos.json`	Currently active shifts	Fresh on page load
`vev_turnos.json`	Live shift tracking	Fresh on refresh
`vev_sesoes_teste.json`	Test sessions	Cached

`vev_laudos.json`	Technical issues	Cached
`vev_abastecimentos.json`	Fuel refueling records	Cached

4.2 Parsers (`app.py:918-1199`)

Function	Lines	Purpose
`parse_hora()`	667	Parse time strings
`parse_turnos_encerrados()`	918	Clean shift data
`parse_abastecimentos()`	981	Normalize fuel records
`parse_sesoes_teste()`	1036	Parse test sessions
`parse_turnos_ativos()`	1069	Parse active shifts
`parse_turnos_live()`	1104	Parse live tracking
`parse_laudos()`	1139	Parse technical issues

4.3 Session State Caching

```
# Keys stored in st.session_state

keys = [
    "last_update",      # Timestamp of last data pull
    "data_loaded",      # Boolean guard
    "data_turnos",      # DataFrame: completed shifts
    "data_abast",       # DataFrame: fuel records
    "data_sesoes",      # DataFrame: test sessions
    "data_laudos",      # DataFrame: laudos
    "data_ativos",      # DataFrame: active shifts
    "data_live",        # DataFrame: live tracking
]
```

On first load, data is fetched from Firebase. On subsequent loads and failures, session state preserves the last valid data.

5 Optimization Patterns

5.1 Error Boundaries

Each tab wraps its content in try/except to prevent a single failure from crashing the entire dashboard.

5.2 Skeleton Loading

CSS shimmer animation on data containers while Firebase responses are pending:

```
@keyframes shimmer { 0%{background-position:-200% 0} 100%{background-position:200% 0} }
.skeleton { background: linear-gradient(90deg, #1e293b 25%, #334155 50%, #1e293b 75%); background-size: 200% 1
```

5.3 Refresh (User-Controlled)

- Button: "Recargar" in sidebar with toast feedback
- Auto-toggle: "Auto-actualizar a cada 30s" (default: off)
- No auto-reload on page load - preserves expanders, scroll, and tab state

5.4 Color Palette

```
CORS = {  
    "ford_blue":    "#002C5B",  
    "accent_blue":  "#1E90FF",  
    "neon_blue":    "#00E5FF",  
    "dark_bg":      "#0B0F17",  
    "card_bg":      "#121824",  
    "border_gray":  "#243042",  
    "green":         "#10B981",  
    "orange":        "#F59E0B",  
    "red":           "#EF4444",  
    "purple":        "#8B5CF6",  
}
```

6 Dependencies

```
streamlit==1.31.0  
  
pandas==2.1.4  
plotly==5.18.0  
gsread==6.0.2  
google-auth==2.26.1  
folium==0.16.0  
streamlit-folium==0.19.0  
openpyxl==3.1.2  
numpy==1.26.3  
scipy==1.12.0  
streamlit-autorefresh==1.0.1
```